

# Marco Linton

## PERSONAL INFORMATION

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## ACADEMIC POSITIONS

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- 2027- Ramon y Cajal Research Fellow, *Instituto de Ciencias Matemáticas (ICMAT)*.  
2024-2027: Severo Ochoa Postdoctoral Research Associate, *Instituto de Ciencias Matemáticas (ICMAT)*.  
2022-2024: Postdoctoral researcher, *University of Oxford*.  
Dawid Kielak's group.

## EDUCATION

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- 2018–2022: PhD in Mathematics: *University of Warwick*.  
Supervised by Saul Schleimer.  
Thesis title: *One-relator hierarchies*.  
2017–2018: MSc in Mathematics, *Imperial College London*.  
First class honours with distinction.  
2014–2017: BSc in Mathematics and Computer Science, *University of York*.  
First class honours with distinction.

## RESEARCH INTERESTS

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Coherence, compact cores, splittings,  $L^2$ -invariants, group cohomology, free groups, one-relator groups, hyperbolic groups, quasi-convex subgroups.

## PREPRINTS

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1. *Finite complete rewriting systems for graphs of free groups with applications to free-by-cyclic, one-relator, and three-manifold groups*, (with Robert Gray), 2026, [arXiv:2607.07496](https://arxiv.org/abs/2607.07496).
2. *Coherent RFRS groups* (Appendix to main article by Sam P. Fisher and Pablo Sánchez-Peralta), 2026, [arXiv:2603.16763](https://arxiv.org/abs/2603.16763).
3. *The finitely generated intersection property in fundamental groups of graphs of groups* (with Jordi Delgado, Jone Lopez de Gamiz Zearra, Mallika Roy and Pascal Weil), 2025, [arXiv:2512.12635](https://arxiv.org/abs/2512.12635).
4. *Group pairs, coherence and Farrell–Jones Conjecture for  $K_0$*  (with Andrei Jaikin-Zapirain and Pablo Sánchez-Peralta), 2025, [arXiv:2510.023518](https://arxiv.org/abs/2510.023518).
5. *The geometry of subgroups of mapping tori of free groups*, 2025, [arXiv:2510.03145](https://arxiv.org/abs/2510.03145).
6. *Pullbacks and intersections in categories of graphs of groups* (with Jordi Delgado, Jone Lopez de Gamiz Zearra, Mallika Roy and Pascal Weil), 2025, [arXiv:2508.04362](https://arxiv.org/abs/2508.04362).
7. *On the intersections of finitely generated subgroups of free groups: reduced rank to full rank*, 2021, [arxiv:2108.10814](https://arxiv.org/abs/2108.10814).

## PUBLICATIONS

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8. *The theory of one-relator groups: history and recent progress*, (with Carl-Fredrik Nyberg-Brodda), 2025, Chapter to appear in Recent Progress in Mathematics, KIAS Springer Ser. Math., 2., available at [arXiv:2501.18306](#).
9. *Lifting relations in right orderable groups*, 2024, Accepted for publication at J. Lond. Math. Soc., available at [arXiv:2412.17057](#).
10. *Embedding finitely generated free-by-cyclic groups in {finitely generated free}-by-cyclic groups*, 2025, *Int. Math. Res. Not.* **2026** (2026), no. 4: [MR5035094](#).
11. *Residually rationally solvable one-relator groups*. *Bull. Lond. Math. Soc.* **58** (2025), no. 1: [MR5019869](#).
12. *Group rings of three-manifold groups* (with Dawid Kielak). *Proc. Amer. Math. Soc.* **152** (2024), no. 5, 1939–1946; [MR4728464](#).
13. *On the coherence of one-relator groups and their group algebras* (with Andrei Jaikin-Zapirain). *Ann. of Math. (2)* **201** (2025), no. 3, 909–959. [MR4899802](#).
14. *Virtually free-by-cyclic groups* (with Dawid Kielak). *Geom. Funct. Anal.* **34** (2024), no. 5, 1580–1608; [MR4792841](#).
15. *Hyperbolic one-relator groups*. *Canad. J. Math.* **78** (2026), no. 1, 35–61. [MR5020065](#).
16. *One-relator hierarchies*. *Duke Math. J.* **174** (2025), no. 4, 747–802. [MR4905535](#).
17. *The fully compressed subgroup membership problem*. *J. Algebra* **628** (2023), 562–583; [MR4579901](#).

## INVITED CONFERENCE TALKS

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| Jul. 2026 | Arithmetic Groups, High-dimensional Expanders and $L^2$ -invariants (Brin Mathematics Research Center): <i>The Hanna Neumann conjecture and submultiplicativity of <math>L^2</math>-Betti numbers.</i> |
| Jun. 2026 | Young Geometric Group Theory XIV (KIAS): <i>Hanna Neumann bounds for cubulated locally quasi-convex hyperbolic groups.</i>                                                                             |
| Jun. 2026 | Sapir distinguished Lecture at GAGTA (CUNY): <i>One-relator groups: past, present and future.</i>                                                                                                      |
| Aug. 2025 | Algebraic and Analytic Methods in Group Theory in Warsaw (IMPAN): <i>Coherence of group pairs.</i>                                                                                                     |
| Aug. 2025 | Operators, Graphs, Groups Colloquium (Newton Institute): <i>The geometry of subgroups of free-by-cyclic groups.</i>                                                                                    |
| Jul. 2025 | JAE School of Mathematics (ICMAT): <i>How complicated can a group with a single defining relation really be?</i>                                                                                       |
| Jun. 2025 | Geometric Group Theory and Related Topics (Peking University): <i>Quasi-convex subgroups of free-by-cyclic groups.</i>                                                                                 |
| Jan. 2025 | RSME's 7th Congress of Young Researchers (University of the Basque country): <i>The geometry of free-by-cyclic groups.</i>                                                                             |
| Sep. 2024 | Combinatorial Non-positive Curvature (Banff International Research Station): <i>Lifting relations in right orderable groups.</i>                                                                       |
| May. 2024 | Barcelona Weekend in Group Theory (Universitat Politècnica de Catalunya): <i>The relation gap and relation lifting problems for groups with cyclic relation modules.</i>                               |

- Apr. 2024 Groups, Topology and Applications in Gran Bilbao V (University of the Basque Country): *The relation gap and relation lifting problems for groups with cyclic relation modules.*
- Feb 2024 GAGTA (CIRM): *The coherence of one-relator groups.*
- Jun. 2023 Groups and Computation (Stevens Institute of Technology): *How to prove one-relator groups are coherent.*
- May 2023 Geometry of subgroups (Centre de Recherches Mathématiques): *The coherence of one-relator groups.*
- May 2023 One-day special NYGT event in memory of Gilbert Baumslag and Benjamin Fine (City University of New York): *One-relator groups, virtually free-by-cyclic groups and coherence.*
- Oct. 2022 Groups and Geometry in the South east (University College London): *Hyperbolic one-relator groups.*
- Sep. 2022 Modern Advances in Geometric Group Theory (University of Manchester): *Simplifying Gersten's conjecture.*
- Aug. 2021 World of groupcraft (online conference): *Fibre products, free groups and regular languages.*

## SEMINAR TALKS

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- May. 2026 Long-distance seminar on Geometric Group Theory in Mexico (Centro de Ciencias Matemáticas): *TBA.*
- Mar. 2026 McGill Geometric Group Theory (McGill University): *Characterising local quasi-convexity.*
- Mar. 2026 Geometry and Topology Seminar (University of Bristol): *Characterising local quasi-convexity.*
- Jan. 2026 Algebra, Geometry and Topology Department Colloquium (Universidad Complutense de Madrid): *The geometry of one-relator groups.*
- Nov. 2025 Geometry Seminar (Max Planck Institute Leipzig): *The geometry of mapping tori of graphs.*
- Nov. 2025 Glasgow Geometry & Topology seminar (University of Glasgow): *Coherence of groups pairs.*
- Apr. 2025 Geometric Group Theory Seminar (University of Bonn): *The geometry of free-by-cyclic groups.*
- Apr. 2025 Madrid Group Theory Seminar (ICMAT): *The geometry of free-by-cyclic groups.*
- Dec. 2024 Wrocław Geometry Seminar (University of Wrocław): *The geometry of free-by-cyclic groups.*
- Nov. 2024 KIAS-Springer Lectures (Korean Institute for Advanced Study): *An overview of recent progress in the theory of one-relator groups.*
- Oct. 2024 BG-UToledo Geometry and Topology Seminar (Bowling green University and University of Toledo): *The relation gap and relation lifting problems.*
- Jul. 2024 Junior Topology and Group Theory Seminar (University of Oxford): *The relation gap and relation lifting problems.*
- Mar. 2024 Géométrie dynamique (Université de Lille): *The coherence of one-relator groups.*
- Mar. 2024 Geometry and Topology Hybrid (University of Warwick): *The coherence of one-relator groups.*
- Mar. 2024 Algebraic Topology Seminar (University of Warwick): *The relation gap conjecture for groups with cyclic relation modules.*
- Feb. 2024 Topology Seminar (Princeton University): *The coherence of one-relator groups.*

- Jan. 2024 Geometric Group Theory Seminar (Univeristy of Bonn): *The coherence of one-relator groups.*
- Nov. 2023 Geometry and Topology Seminar (University of Bristol): *The coherence of one-relator groups.*
- Nov. 2023 Group Theory Seminar (Instituto Ciencias Matemáticas): *Virtually free-by-cyclic groups, one-relator groups and coherence.*
- Nov. 2023 Group Theory Seminar (Universitat Politecnica de Catalunya): *The coherence of one-relator groups.*
- Oct. 2023 Purdue Geometry and Geometric Analysis Seminar (Purdue University): *Coherence, non-positive immersions and  $L^2$ -Betti numbers.*
- Oct. 2023 Manchester Algebra Seminar (University of Manchester): *Coherence, non-positive immersions and  $L^2$ -Betti numbers.*
- Mar. 2023 Geometry and Analysis on Groups Research Seminar (Universität Wien): *Virtually free-by-cyclic groups.*
- Dec. 2022 Algebra, Geometry and Topology Seminar (Heriot-Watt University): *Hyperbolic one-relator groups.*
- Nov. 2022 Geometric Group Theory Seminar (University of Oxford): *Hyperbolic one-relator groups.*
- Nov. 2022 Fudan Topology Seminar (Fudan University): *Hyperbolic one-relator groups.*
- Oct. 2022 Dagger seminar (University of Warwick): *Posion subgroups for hyperbolic groups.*
- May 2022 Pure Postgraduate Seminar (University of Southampton): *Poison subgroups. for hyperbolic groups.*
- Apr. 2022 Geometry and topology online (University of Warwick): *Hyperbolicity of certain one-relator groups.*
- Apr. 2022 Algebra seminar (University of Bristol): *Hyperbolicity of certain one-relator groups.*
- Apr. 2022 New York Group Theory Seminar (City University of New York): *Hyperbolicity of certain one-relator groups.*
- Apr. 2022 Geometry and Analysis on Groups Research Seminar (Universität Wien): *Primitivity rank, one-relator groups and hyperbolicity.*
- Mar. 2022 Group theory seminari (Universitat Politecnica de Catalunya): *Primitivity rank, one-relator groups and hyperbolicity*
- Nov. 2021 Geometric group theory seminar (University of Cambridge): *When does a one-relator group have a quasiconvex hierarchy?*
- Oct. 2021 Dagger seminar (University of Warwick): *Dynamics and isoperimetric inequalities for the double mapping torus.*
- Apr. 2021 Pure PGR seminar (University of Southampton): *An ode to the fibre product graph.*
- Jan. 2020 Bridges seminar (University of Bristol): *Hierarchies for one-relator groups.*
- Dec. 2019 Junior geometry and topology seminar (University of Oxford): *Hierarchies for one-relator groups.*
- Dec. 2019 Postgraduate seminar (University of Warwick): *Hierarchies for one-relator groups.*
- Nov. 2019 Dagger seminar (University of Warwick): *Hierarchies for one-relator groups.*
- Feb. 2019 Gateway seminar (University of Warwick): *Dehn filling in relatively hyperbolic groups with torsion.*

## MINICOURSES

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- Sep. 2027 I will give a minicourse at the Ventotene International Workshop ‘Negatively Curved Immersions, Coherence and the Surface Subgroup Conjecture’.
- Oct. 2026 I will give an 8 hour minicourse at the Shanghai Center for Mathematical Sciences on *Free-by-cyclic groups and their subgroups*.
- Jun. 2026 I gave a 3 hour minicourse at ‘Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future’ at the ICMAT on: *2-complexes with vanishing  $L^2$ -Betti numbers*.
- Nov. 2024 I gave a 3 hour minicourse at the ‘KIAS workshop on One Relator groups and other Aspects of GGT (KOREA-GGT)’ on: *One-relator groups and the coherence property*.
- Jul. 2024 I gave a 9 hour graduate course at the ‘Topological and analytic methods in group theory’ conference at the University of Hawaii on: *Coherence, homological coherence and one-relator groups*.
- Apr. 2024 I gave an 8 hour minicourse to experts at the conference ‘Groups, Topology and Applications in Gran Bilbao V’ at the University of the Basque country on: *The coherence of one-relator groups*.

## SUPERVISION

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- Alba Gonzalez Gonzalez (Masters student at Universidad Autónoma de Madrid), now pursuing a PhD at Oxford.
- Miguel Angel Royo Miguel (JAE summer school research student from Universidad de Zaragoza).

## PANELS AND COMMITTEES

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- Director of the thesis committee for Pablo Sánchez Peralta’s PhD defence (Universidad Autónoma de Madrid).
- Scientific committee for the conference ‘Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future.’
- Academic career panel at conference ‘Topological and analytic methods in group theory.’
- Committee for Sam Fisher’s confirmation of status (University of Oxford).

## TEACHING

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In 2025 I taught the advanced algebra course at the Universidad Autonoma de Madrid (UAM) on *Geometric Group Theory*.

## SUPPORT CLASSES

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Between 2018 and 2022 I worked as a teaching assistant for the University of Warwick. I provided support classes and marked assignments for the following courses:

- Discrete Mathematics and its Applications I.
- Algebra II.
- Geometry.

- Introduction to Topology.
- Algebraic Topology.

## AWARDS AND GRANTS

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- Awarded a Ramon y Cajal grant (2025). Ranked number 1 in mathematical sciences.
- Awarded the Sapir Distinguished Lecture (2025).
- LMS research in pairs grant to visit Dr. Robert Gray (2023).
- Departmental funding for supervision of undergraduate summer project, *Finding solutions to the Post correspondence problem for free groups*, University of Oxford (2023).
- Winner of the prize for best MSc pure mathematics student, Imperial College London (2018).
- Winner of the Kathleen Ryan project prize for the best overall final year BSc project performance, University of York (2017).

## ORGANISATIONAL ACTIVITIES

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- Organiser for the conference *Young Geometric Group Theory XV* at the ICMAT in 2027.
- Organiser for the conference *Groups in Madrid* at the ICMAT in 2025.
- Organiser for the Dynamics and Group Geometry Early Researchers (DAGGER) seminar at the University of Warwick.
- Organiser for a reading group on 3-manifolds, following *An introduction to geometric topology* by Bruno Martelli.
- Organiser for a reading group on CAT(0) cube complexes, following *The structure of groups with a quasi-convex hierarchy* by Dani Wise.
- Organiser for a reading group on JSJ decompositions, following *JSJ decompositions of groups* by Vincent Guirardel and Gilbert Levitt.

## LANGUAGES

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- English: mothertongue
- Italian: fluent.
- Spanish: working knowledge.